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Received: 19 May 2025

Accepted: 12 December 2025

Published online: 28 December 2025

Cite this article as: Huan J., Li M. & Zhou H. Emotion-aware adaptation of CLIP model for facial expression recognition. *Artif Intell Rev* (2025). <https://doi.org/10.1007/s10462-025-11468-4>

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Emotion-aware Adaptation of CLIP model for Facial Expression Recognition

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Abstract

Facial Expression Recognition (FER) remains a challenging task due to subtle variations in facial details and unconstrained conditions such as changes in head posture, illumination, and occlusion. Current FER approaches primarily focus on capturing discriminative facial features in vision manner, often neglecting the rich semantic information available in textual modalities. Additionally, these methods typically rely on generic classification templates, which fail to capture instance-specific features, resulting in inadequate representation and fine-grained discrimination ability. To tackle the above issues, we propose a novel emotion-aware adaptation framework that integrates the pre-trained CLIP model for FER, leveraging both visual and textual modalities to enhance representation learning and capture fine-grained emotional details. Specifically, we introduce the Expression-aware Adapter (EAA) module to capture emotion-specific facial representations through task-specific fine-tuning while preserving the generalization capabilities of the CLIP model. Furthermore, the Instance-enhanced Expression Classifier (IEC) module is proposed to enhance textual descriptors with instance-specific visual embeddings using spherical linear interpolation, creating a more precise and discriminative classifier. Extensive experiments on three in-the-wild FER benchmarks demonstrate superiority of our proposed approach.

Keywords: Facial Expression Recognition; Vision-Language Pre-training Model; Visual Adapter; Spherical Linear Interpolation

1 Introduction

Facial expressions are crucial nonverbal cues to convey human emotions, making their accurate recognition highly significant for various applications, including human-computer interaction [1], healthcare services [2], and fatigue driving detection [3]. Due to the subtle variations in facial details, facial expression recognition (FER) remains a challenging task, requiring a series of steps such as data preprocessing, feature extraction, and classification.

Traditional hand-crafted feature extraction methods, such as LBP [4], HOG [5], and SIFT [6], have shown promising performance on controlled in-the-lab datasets like CK+ [7] and JAFFE [8]. However, these methods often fall short in capturing the complexity and diversity of facial expressions, particularly under variations in head posture, illumination, and occlusion, thus limiting their effectiveness in real-world scenarios.

With the advancement of deep learning techniques, large-scale in-the-wild datasets, such as AffectNet [9], RAF-DB [10], and ExpW [11], have been introduced, presenting more challenging recognition tasks. Leveraging these in-the-wild FER datasets, researchers have developed various deep learning-based methods to enhance FER accuracy in diverse real-world scenarios [12–15]. To extract the rich visual information from facial images, a range of local attention-based FER methods [12, 16–18] has been developed, applying Convolutional Neural Networks (CNNs) layer by layer to effectively capture local features. However, these methods often overlook broader contextual relationships, such as global facial structures and inter-region dependencies. To address these limitations, a series of global perception-based methods [15, 19–21] have emerged, exemplified by the Vision Transformer (ViT) [22] with their self-attention mechanism, capturing global dependencies and contextual information. Despite these advancements, the aforementioned methods are predominantly unimodal, focusing solely on visual information and neglecting the potential benefits of incorporating textual modalities. This limitation highlights the need for more comprehensive approaches that leverage multi-modal information to further improve FER performance.

Recently, Vision-Language Pre-training (VLP) methods, such as Contrastive Language-Image Pretraining (CLIP) [23], have demonstrated remarkable capabilities in integrating visual and textual modalities using large-scale image-text pair datasets. By jointly processing visual and textual information, CLIP creates a unified representation space that effectively captures the complex interactions between images and language, enabling impressive generalization across diverse tasks [24, 25]. Building on these advancements, recent studies have explored the application of transfer learning from VLP models to various downstream tasks, such as video understanding [26], semantic segmentation [27], and object detection [28]. Additionally, research has shown that VLP models can learn rich expression-related features [14, 29], making them promising for FER tasks. However, applying VLP models to FER tasks presents two major challenges that must be addressed to fully leverage their potential.

First, the CLIP model tends to be biased towards macro-objects (*e.g.*, the entire head and face), resulting in micro-objects (*e.g.*, muscle) containing critical information being represented further from the textual representation than the background or macro-objects. This bias arises since CLIP is typically trained on coarse-grained

natural images, where dominant pixels from large-scale, easily distinguishable elements attract more of the model’s focus [30]. Nonetheless, facial expressions such as “happy” and “surprise” are abstract and nuanced, involving intricate muscle movements, lighting, and cultural influences, which the model’s object-centric manner fails to adequately capture. Therefore, reliance on the current CLIP model is inadequate for accurately representing and comprehending facial expressions. In addition, the general classifier with the template “a photo of [class]” (*e.g.*, happy) in the CLIP model neglects to consider the distinctive features and subtleties inherent in facial expressions. Although several expression-specific prompts [14, 29], such as “a photo of a [class] expression, a type of facial expression”, have been designed for FER tasks, they still neglect the inherent instance-specific information in the images. In fact, each facial expression image has a unique combination of factors, including facial muscle movements, lighting conditions, and the individual’s facial structure. The general classifier’s prompts are overly abstract and cannot capture these specific details, leading to a lack of fine-grained discrimination ability.

To address the challenges outlined above, we propose a novel Emotion-aware Adaptation framework named **EA-CLIP**, which integrates the pre-trained CLIP model with two innovative modules: the Expression-aware Adapter (EAA) and the Instance-enhanced Expression Classifier (IEC). Our EA-CLIP leverages both visual and textual modalities while preserving the nuanced relationships between image and text data, thereby enhancing the accuracy and robustness of FER. Specifically, the EAA module is designed to capture fine-grained, emotion-specific facial representations by introducing a lightweight, task-specific fine-tuning manner that retains the generalization capabilities of the CLIP model. Meanwhile, the IEC module enhances the expressiveness of textual descriptors by incorporating instance-specific visual embeddings, using spherical linear interpolation to create a more precise and discriminative classifier. By combining the EAA and IEC modules, our approach effectively represents and comprehends facial expressions, improving specificity and discrimination capabilities.

The main contributions of this work can be summarized as follows:

- 1) We propose an emotion-aware adaptation framework, named EA-CLIP, which integrates the pre-trained CLIP model for FER. This framework effectively combines visual and textual modalities to enhance representation learning, allowing for the capture of fine-grained emotional details.
- 2) To identify emotion-related facial variations, we introduce the Expression-aware Adapter (EAA) module, which captures discriminative and emotion-specific visual representations through lightweight, task-specific fine-tuning. Additionally, the Instance-enhanced Expression Classifier (IEC) module is designed to incorporate instance-specific visual embeddings into textual descriptors using spherical linear interpolation, enhancing the discriminative FER classifier.
- 3) Extensive experimental results on three in-the-wild FER benchmarks demonstrate the superiority of the proposed approach.

2 Related Work

In this section, related works about Facial Expression Recognition, Vision-Language Pre-training and Transfer Learning are reviewed briefly.

2.1 Facial Expression Recognition

In recent decades, Facial Expression Recognition (FER) has garnered significant attention and research due to its critical role in various applications [31]. Substantial advancements have been achieved, evolving from traditional methods to modern deep-learning techniques. Early researches primarily rely on hand-crafted features or shallow learning techniques, such as LBP [4], HOG [5] and SIFT [6]. These approaches were designed to capture texture, edge, and gradient information relevant to facial expressions through specifically engineered features. However, their performance was heavily dependent on expert knowledge and proved inadequate when handling complex scenarios, such as lighting variations and occlusions.

Building on the remarkable success of deep neural networks in various computer vision tasks [32–36], researchers have developed numerous deep-learning methods to enhance FER performance [11, 13–15, 20, 29, 37, 38]. One prominent category of these methods is based on Convolutional Neural Networks (CNNs) [12, 17, 18, 37–44]. For instance, Wang et al. [37] randomly crop multiple facial local patches and apply an attention network to identify and focus on the salient regions. Similarly, Li et al. [41] propose a coarse-to-fine label strategy to enable the model to classify highly similar representations more effectively. Ruan et al. [45] reconstruct decomposed facial action-aware features into expression-specific features, successfully capturing expression similarities and unique variations. Besides, Zeng et al. [38] employ the meta-learning strategy to address class imbalance and distribution mismatch, significantly improving FER performance. Liu et al. [16] emphasize the importance of extracting features from different facial regions to capture diverse and critical emotional information. Additionally, Li et al. [17] implements a hierarchical attention strategy to extract local fine-grained features while incorporating a graph network to model global relationships.

Inspired by Transformer [46], Vision Transformer (ViT) [22] utilizes a self-attention mechanism to capture global information within images and has achieved significant success in image classification tasks. Building upon this foundation, another series of methods focuses on ViT-based FER [15, 19–21, 47–49] have been proposed, demonstrating promising performance. Among them, Ma et al. [19] integrate feature maps generated by a dual-branch CNN network and feeds them into ViT to explore relationships between visual tokens. Similarly, Xue et al. [20] deploy an multi-head attention and dropping module to adaptively learn multiple local patches and utilize a visual transformer to capture relationships between them. Building on this, Xue et al. [15] employ a pooling scheme to explicitly reduce input tokens before feeding them into ViT, enhancing recognition accuracy. Moreover, Huang et al. [48] introduce double attention mechanisms to simultaneously capture dependencies between different regions and learn global representations. Although the aforementioned methods have demonstrated strong performance, they predominantly rely on visual encoders for

feature extraction, overlooking the rich semantic information provided by text. In contrast, we leverage the CLIP model to jointly model facial images and emotion-related textual descriptions, facilitating the extraction of both general and expression-specific visual representations. Additionally, we incorporate instance-specific visual information into the expression classifier, further enhancing the model’s discriminability for FER.

2.2 Vision-Language Pre-training and Transfer Learning

In recent years, inspired by the success of pre-trained models in the fields of Computer Vision (CV) [22, 50] and Natural Language Processing (NLP) [46, 51], Vision-Language Pre-training (VLP) models have received increasing attention in the CV community. VLP models like CLIP [23], ALIGN [25], DeCLIP [52], and CLIPPO [53], pre-trained on large-scale image-text pairs, have shown promising generalization performance. By leveraging larger amounts of multi-modal data, these models demonstrate promising performance across a variety of downstream applications.

To transfer pre-trained VLP models to downstream tasks, various parameter-efficient methods have been proposed in the NLP and CV communities, such as image classification [54], object detection [55], and semantic segmentation [56]. These approaches can be broadly classified into token-based prompt learning (prompting) and network-based adaptation (adapters). For prompting, it typically begins by providing textual instructions to the language component of VLMs. For example, CoOp [57] improves CLIP’s performance in few-shot learning by replacing hand-crafted prompts with a learnable set of continuous vectors in its language branch. Building on this, CoCoOp [58] extends CoOp by conditioning the prompts on specific image instances, enhancing generalization to unseen classes. Additionally, studies such as VPT [59] and Visual Prompting [60] explore the use of prompting in the vision branch of VLMs. In contrast, adapter-based methods often rely on uni-modal adapters for fine-tuning. For instance, Clip-Adapter [61] and Tip-Adapter [62] introduce adapter layers after the image encoder to refine performance. However, these methods predominantly focus on macroscopic objects with an object-centric bias [30, 63], which limits their ability to generate fine-grained textual descriptors for facial details.

More recently, VLP models have been successfully applied to FER [14, 29, 64–66]. For example, CLIPER [14] leverages the pre-trained CLIP model alongside multiple text descriptors to extract expression-related representations. Similarly, CEPrompt [29] adopts an emotion-aware prompting strategy to capture category-specific appearance features, further enhancing the representation of facial expressions. In Video FER, DFER-CLIP [66] incorporates a transformer-based module to capture temporal information. However, it requires fully fine-tuning the image encoder, which leads to inefficiency. Subsequently, EmoCLIP [65] introduces a novel vision-language model that uses sample-level text descriptions as natural language supervision for Video FER. Nevertheless, the above methods overlook the inherent instance-specific information present in images. In contrast, we integrate instance-specific image embeddings into the textual descriptors, allowing for a more nuanced and precise representation of each image’s unique characteristics.

3 Proposed method

In this section, we first provide a brief revisit of Contrastive Language-Image Pre-training. After that, we introduce the proposed EA-CLIP, including the Expression-aware Adapter and the Instance-enhanced Expression Classifier.

3.1 Preliminaries

3.1.1 Contrastive Language-Image Pre-training

Model and Training: Inspired by advances in vision-language pre-training (VLP) models [23, 25, 53], Contrastive Language-Image Pre-training (CLIP) [23] has emerged as a foundational framework for aligning visual and textual embeddings in a shared representation space. CLIP is trained on web-scale datasets with 400 million image-text caption pairs. It consists of an image encoder $\mathbf{E}^I$ and a text encoder $\mathbf{E}^T$ to produce visual and textual embeddings in the same representation space, to pull related image-text pairs closely, and vice versa for unrelated pairs. By this large-scale pre-training, CLIP can simultaneously encode images and text descriptions to perform a wide range of downstream tasks.

Inference: Since CLIP aligns textual and visual embeddings within a common representation space, allowing inference to be based on their cosine similarity. Specifically, CLIP employs a standard template $\mathbf{p}$ (e.g., “a photo of [class]”) to generate text descriptors for each class. The prediction probability for an input image $\mathbf{x}$ belonging to the c -th class is calculated using this alignment between visual embeddings $\mathbf{v}=\mathbf{E}^I(\mathbf{x})$ and textual embeddings $\mathbf{w}=\mathbf{E}^T(\mathbf{p})$, which can be formulated as:

$$p(y = c|\mathbf{x}) = \frac{\exp(\cos(\mathbf{v}, \mathbf{w}_c)/\tau)}{\sum_{i=1}^{|C|} \exp(\cos(\mathbf{v}, \mathbf{w}_i)/\tau)}. \quad (1)$$

Here, $\cos(\cdot, \cdot)$ represents the cosine similarity, τ is a temperature parameter, and C denotes the total number of classes.

3.2 EA-CLIP

The overview framework of our proposed EA-CLIP is illustrated in Fig. 1. Specifically, to capture emotion-related discriminative representations in facial images, we introduce the Expression-aware Adapter (EAA) module, which focuses on fine-grained, emotion-specific facial details. Given a facial image $\mathbf{x}$, the EAA module extracts emotion-aware representations through task-specific fine-tuning, maintaining the generalization capabilities of the pre-trained CLIP model. Meanwhile, the expression labels are combined with textual descriptors and fed into a text encoder to obtain textual features. Subsequently, these representations are enhanced using the Instance-enhanced Expression Classifier (IEC) module, which incorporates instance-specific visual embeddings into textual descriptors via spherical linear interpolation, creating a more precise and discriminative classifier. Our EA-CLIP significantly improves the model’s ability to capture emotion-related discriminative representations, enhancing overall performance and reliability in FER tasks.

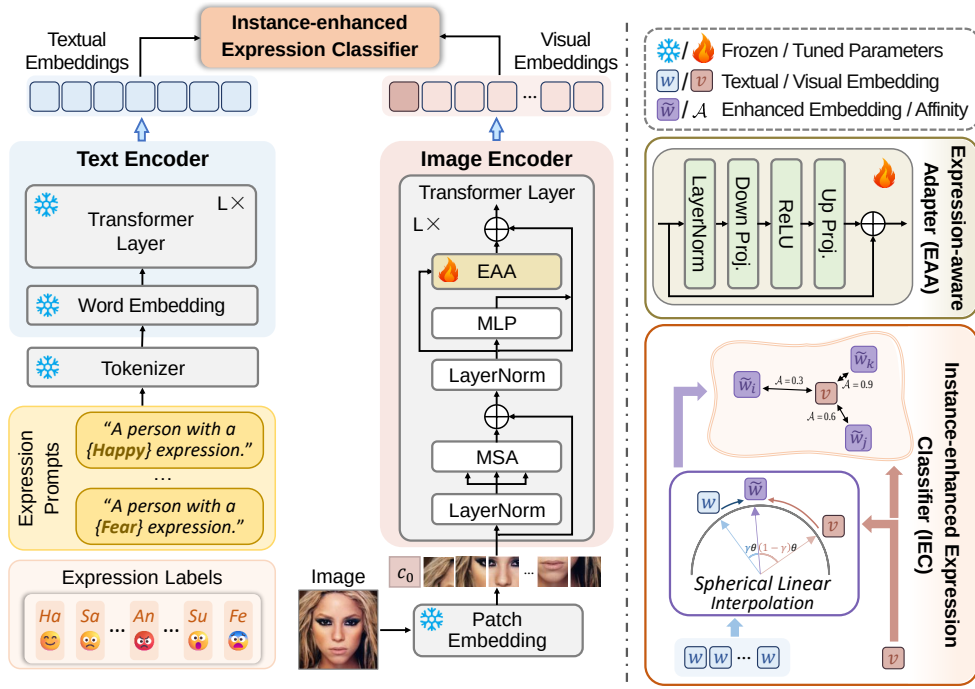


Fig. 1: The overview of our proposed EA-CLIP, which consists of two key modules: Expression-aware Adapter (EAA) to extract emotion-specific facial representations, and Instance-enhanced Expression Classifier (IEC) to enhance the expressiveness of the textual embeddings for classification.

3.2.1 Visual and Textual Embedding

Given a facial expression dataset $\mathcal{D} = \{(\mathbf{x}_n, \mathbf{t}_n)\}_{n=1}^N$, where $\mathbf{x}_n \in \mathbb{R}^{H \times W \times 3}$ is an image, H and W are the height and width, while $\mathbf{t}_n$ is a tokenized text expression category (e.g., happy) of n -th sample. For simplicity, we omit the subscripts for each sample. Firstly, an image $\mathbf{x}$ will be fed into the image encoder E_I to obtain the image feature $\mathbf{v}$ as follows:

$$\begin{aligned} \mathbf{v}_0 &= \text{PatchEmbed}(\mathbf{x}), \\ [\mathbf{c}_l, \mathbf{v}_l] &= \mathbf{E}_l^I([\mathbf{c}_{l-1}, \mathbf{v}_{l-1}]) \quad l = 1, 2, \dots, L, \\ \mathbf{v} &= \text{PatchProj}(\mathbf{c}_L). \end{aligned} \quad (2)$$

As shown in Fig. 1, the *PatchEmbed* layer first splits the input image x into fixed-size patches. Then these patches are projected into embeddings $\mathbf{v}_0$, concatenated with a learnable class token $\mathbf{c}_0$ to form embeddings $[\mathbf{c}_0, \mathbf{v}_0]$. After that, the concatenated embeddings are sequentially passed through L transformer blocks $\{\mathbf{E}_l^I\}_{l=1}^L$. The Transformer block [46] consists of alternating layers of multi-headed self-attention

(MSA) and MLP blocks. LayerNorm (LN) is applied before every block, and residual connections [67] after every block. For clarity, we define an intermediate variable $\mathbf{z}$ representing the concatenation of embeddings $\mathbf{c}$ and $\mathbf{v}$. The feedforward process of each transformer block is as follows:

$$\begin{aligned} \mathbf{z}_{l-1} &= [\mathbf{c}_{l-1}, \mathbf{v}_{l-1}] \quad l = 1, 2, \dots, L, \\ \mathbf{z}'_l &= \text{MSA}(\text{LN}(\mathbf{z}_{l-1})) + \mathbf{z}_{l-1} \quad l = 1, 2, \dots, L, \\ \mathbf{z}_l &= \text{MLP}(\text{LN}(\mathbf{z}'_l)) + \mathbf{z}'_l \quad l = 1, 2, \dots, L. \end{aligned} \quad (3)$$

Finally, a *PatchProj* layer projects the class token $\mathbf{c}_L$ of the last transformer block $\mathbf{E}_L^I$ to the image feature $\mathbf{v}$ into the common vision-language space.

Simultaneously, given an expression category $\mathbf{t}_c$ ($c = 1, 2, \dots, C$), where C denotes the number of categories, a manually designed text prompt $\mathbf{p}_c$ (we simplify to $\mathbf{p}$) is constructed as “a person with a $[class_c]$ expression”. Likewise, the text prompt $\mathbf{p}$ will be fed into the text encoder $\mathbf{E}^T$ to obtain textual feature $\mathbf{w}_c$ as follows:

$$\begin{aligned} [\mathbf{w}_0^j]_{j=1}^{N_p} &= \text{TextEmbed}(\mathbf{p}), \\ [\mathbf{w}_l^j]_{j=1}^{N_p} &= \mathbf{E}_l^T([\mathbf{w}_{l-1}^j]_{j=1}^{N_p}) \quad l = 1, 2, \dots, L, \\ \mathbf{w} &= \text{TextProj}(\mathbf{w}_L^{N_p}). \end{aligned} \quad (4)$$

Here, the *TextEmbed* layer is applied to tokenize and project the input text prompt to N_p word embeddings. Then, a series of transformer blocks $\{\mathbf{E}_l^T\}_{l=1}^L$ is used to extract features, and the *TextProj* layer projects the last token $\mathbf{w}_L^{N_p}$ of the last transformer block $\mathbf{E}_L^T$ into the common vision-language space.

3.2.2 EAA: Expression-aware Adapter

The CLIP model has been extensively utilized in various downstream tasks due to its vision-language pre-training, which offers remarkable generalization capabilities [24, 57]. However, when applied to facial expression recognition, it tends to overlook fine-grained, emotion-specific facial representations, instead prioritizing coarse-grained features.

Prior research has shown that fully fine-tuning CLIP requires a large number of trainable parameters, which not only increases the risk of severe overfitting but also undermines its generalization performance [29, 68]. To address these limitations, we draw inspiration from parameter-efficient transfer learning methods [61, 62, 68] and propose a lightweight module for task-specific fine-tuning. Specifically, as shown in Fig. 1(a), the Expression-aware Adapter $\{\text{EAA}\}_{l=1}^L$ includes a down-projection layer $\mathbf{W}_{\text{down}} \in \mathbb{R}^{d \times r}$, an up-projection layer $\mathbf{W}_{\text{up}} \in \mathbb{R}^{r \times d}$, where r and d is the middle dimension, and output dimension of the bottleneck that satisfies $r \ll d$. Additionally, there is a ReLU layer between these projection layers for non-linear property. The adapter consists of the above bottlenecks that contain few parameters relative to the attention and feedforward layers in the original model. For a specific input embedding $\mathbf{z}'_l$ (described in Eq. 3), the EAA module produces the adapted embeddings $\tilde{\mathbf{z}}_l$, which

can be formulated as:

$$\tilde{\mathbf{z}}_l = \text{ReLU}(\text{LN}(\mathbf{z}'_l) \cdot \mathbf{W}_{\text{down}}) \cdot \mathbf{W}_{\text{up}}. \quad (5)$$

After that, the final visual embedding $\mathbf{z}_l$ is adjusted by fusing both the embeddings $\tilde{\mathbf{z}}_l$ and $\mathbf{z}'_l$ with residual connection as:

$$\mathbf{z}_l = \text{MLP}(\text{LN}(\mathbf{z}'_l)) + \mathbf{z}'_l + \alpha \cdot \tilde{\mathbf{z}}_l, \quad (6)$$

where α is a coefficient to balance between expression-specific knowledge and general pre-trained knowledge. Here, $\alpha=0$ denotes the original transformer block without integrating any expression-specific knowledge. In this way, the adapter tends to focus on discriminative visual embeddings while preserving the universal knowledge in CLIP.

3.2.3 Visual-Textual Affinity Matching

Through the aforementioned expression-related visual and textual semantic information mining process, we obtain: 1) visual embedding $\mathbf{v}$ for sample $\mathbf{x}$, and 2) textual embeddings $\{\mathbf{w}_c\}_{c=1}^C$ for C expression categories. We then compute the affinity matrices between $\mathbf{v}$ and $\{\mathbf{w}_c\}_{c=1}^C$, as follows:

$$\mathcal{A}^c(y=c | \mathbf{x}) = \frac{\exp(\cos(\mathbf{v}, \mathbf{w}_c)/\tau)}{\sum_{j=1}^C \exp(\cos(\mathbf{v}, \mathbf{w}_j)/\tau)}, \quad (7)$$

where $\cos(\cdot, \cdot)$ represents the cosine similarity, and τ is a temperature parameter that controls the scaling, similar to Eq. 1. A higher affinity value signifies a stronger alignment between the visual and textual embeddings, indicating a greater likelihood that the input image belongs to a particular class. Then, $\mathcal{A}^c$ is normalized into probabilities using a softmax function and optimized through the standard cross-entropy (CE) loss, formulated as follows:

$$\begin{aligned} \mathcal{L}_{\text{CE}} &= -\frac{1}{\mathcal{N}} \sum_{i=1}^{\mathcal{N}} \sum_{j=1}^C y_i^j \cdot \log(p_i^j), \\ p^j &= \text{softmax}(\mathcal{A}^j), \end{aligned} \quad (8)$$

where, $\mathcal{N}$ and C denote the total number of samples and categories, respectively.

3.2.4 IEC: Instance-enhanced Expression Classifier

As mentioned in section 3.2.3 (*i.e.*, Eq. 7 and 8), for an individual sample $\mathbf{x}$, we compute the similarity between its visual embedding $\mathbf{v}$ and the textual embeddings $\{\mathbf{w}_c\}_{c=1}^C$ of all classes to form the affinity matrices, which is then used for optimization. However, this method typically relies on generic textual descriptors (*i.e.*, “a person with a $[class_c]$ expression”), which may not adequately capture the unique

characteristics of each image. Consequently this oversight can lead to suboptimal classification performance due to the lack of specificity needed to distinguish between similar expressions.

To this end, we propose an Instance-enhanced Expression Classifier (IEC) that integrates instance-specific image embeddings into the textual descriptors using spherical linear interpolation, which enhances the expressiveness of the textual embeddings, allowing for more precise classification. Specifically, we fuse the visual embedding $\mathbf{v}$ and the textual embedding $\mathbf{w}$ using spherical linear interpolation, as follows:

$$\tilde{\mathbf{w}} : \text{Slerp}(\mathbf{v}, \mathbf{w}; \gamma) = \frac{\sin((1 - \gamma)\theta)}{\sin(\theta)} \mathbf{v} + \frac{\sin(\gamma\theta)}{\sin(\theta)} \mathbf{w}. \quad (9)$$

Here, θ is the angle between $\mathbf{v}$ and $\mathbf{w}$, *i.e.*, $\mathbf{v} \cdot \mathbf{w} = \cos(\theta)$. The angle factor γ is a balancing scalar value within the range of $[0, 1]$ that determines the scale of the linear combination between $\mathbf{v}$ and $\mathbf{w}$ to produce the instance-enhanced textual embedding $\tilde{\mathbf{w}}$ as the expression classifier.

The obtained $\tilde{\mathbf{w}}$ adeptly combines the strengths of both image and text modalities, allowing the classifier to leverage detailed, instance-specific features that are essential for distinguishing subtle variations in expressions. By dynamically adjusting the contributions of visual and textual features, our approach significantly enhances the model’s ability to generalize across diverse instances while maintaining high specificity.

3.2.5 Training and Inference

Fine-tuning. During the fine-tuning phase, we selectively optimize the newly introduced parameters in the EAA, while keeping the rest fixed. In practice, to facilitate the rapid adaptation of the pre-trained CLIP model to the FER task, we also fine-tune certain portions of its parameters. Specifically, the original model components (indicated by pink blocks in Fig. 1) load weights from the pre-trained checkpoint, with most parameters remaining frozen, except for the last two layers of the visual encoder. Meanwhile, the newly added parameters (depicted as grey-green blocks) are updated based on the specific data domain using task-specific losses. We compute the affinity matrices between the visual embedding $\mathbf{v}$ and the instance-enhanced textual embedding $\tilde{\mathbf{w}}$, convert them into probability distributions, and optimize them using Eq. 8.

Inference. After fine-tuning, we load the fine-tuned model to extract features from new input data from the test set. We then compute the affinity matrices (similarity) between the extracted visual features and the textual embeddings. This affinity serves as the classification probability, which is subsequently used to perform FER-specific predictions, as described in Eq. 1.

4 Experiments

In this section, we first describe the datasets used for evaluations and experimental settings. Subsequently, we compare the proposed EA-CLIP with state-of-the-art FER

Table 1: The detailed information of compared FER datasets.

Dataset	Train	Test	Class
RAF-DB [10]	12,271	3,068	7
AffectNet [9]	280,000	3,500	8
ExpW [11]	68,845	13,769	7
Pose-RAF-DB [37]	-	1,806	7
Pose-AffectNet [37]	-	2,907	8
Occlusion-RAF-DB [37]	-	735	7
Occlusion-AffectNet [37]	-	683	8

methods and conduct ablation studies to analyze the effectiveness of each module. Finally, we exhibit several attention visualization analysis of our approach.

4.1 Datasets

To comprehensively verify the effectiveness of our proposed EA-CLIP, we consider several widely employed and publicly available in-the-wild datasets, including RAF-DB [10], AffectNet [9], and ExpW [11], along with occlusion and pose-changes test sets as our target datasets for FER. The detailed information of each dataset is shown in Table 1 and as follows:

- **RAF-DB** [10] consists of 15,339 real-world facial images labeled with seven fundamental expressions: anger, disgust, fear, happy, neutral, sad, and surprise. Following prior works [12], 12,271 images are allocated for training, while the remaining 3,068 are used for evaluation in our experiment.
- **AffectNet** [9] is one of the largest publicly available FER datasets, containing nearly one million facial images collected from the internet. Here, approximately 450,000 images are manually annotated with categorical emotion labels. Following [15], we utilize a subset of 280,000 images for training and 4,000 images for validation, covering eight categories: the seven found in RAF-DB, and “contempt”. The dataset is known for its class imbalance and diverse image quality, making it a challenging benchmark for FER models.
- **ExpW** [11] is a large-scale FER dataset constructed using images retrieved from Google Search. The dataset includes annotations for seven emotion categories, consistent with RAF-DB. Based on the commonly adopted split [15, 69], the dataset comprises 68,845 images for training, 9,179 for validation, and 13,769 for testing.
- **Pose-RAF-DB and Pose-AffectNet** [37] are subsets of RAF-DB and AffectNet, respectively, designed to assess model performance on images with significant facial pose variations. These subsets contain samples with facial pose angle. Specifically, Pose-RAF-DB includes 1,248 images with poses $\geq 30^\circ$ and 558 images with poses $\geq 45^\circ$, while Pose-AffectNet comprises 1,949 and 958 images for the same pose thresholds.
- **Occlusion-RAF-DB and Occlusion-AffectNet** [37] consist of images with occluded faces, drawn from the test set of RAF-DB and the validation set

of AffectNet, respectively. These datasets contain 735 and 683 images, where occlusions stem from objects, hands, or accessories. For RAF-DB and ExpW, the overall accuracy on the test set is used for measurement, while the mean class accuracy on the validation set is reported for AffectNet, following the settings in [15, 29, 37].

4.2 Experimental Setting

All the experiments of our EA-CLIP are built on a CLIP model with the backbone of ViT-B/16 using a single NVIDIA RTX 3090 GPU with 24GB of memory for fairness and consistency. For data processing, all images are resized to 224×224 . We employ the CLIP-ViT-B/16 model for our backbone. Specifically, the image encoder is a ViT network with 12-layer Transformer [46] blocks, while the text encoder $f_t(\cdot)$ here is also a Transformer network with architectural modifications as described in [70]. Following [23], we employ 12-layer Transformer blocks ($L=12$) with 8 attention heads and cap the maximum sequence length of each input text at 76. Correspondingly, we insert an Expression-aware Adapter (EAA) into each Transformer block of the image encoder. Note that all the parameters in the EAAs and the last two transformer blocks in the image encoder are trainable, while the remaining layers are kept fixed. Additionally, following previous work [14, 29], we freeze the entire text encoder. During training on RAF-DB and ExpW, the batch size is set to 128, and the model is trained for 34 epochs using the AdamW optimizer with an initial learning rate of $1e-4$. For the large-scale dataset AffectNet, the model is trained for 4 epochs with a batch size of 256 and an initial learning rate of $1e-5$. Furthermore, grid search is applied to determine all hyper-parameters. Specifically, the adapter factor α and the angle factor γ are set to 0.5 and 0.7 for AffectNet, and 0.7 and 0.9 for the remaining datasets, respectively. The temperature factor τ is set to 0.7 across all datasets. We implement our proposed method based on Pytorch 1.13.1.

4.3 Comparison with state-of-the-art

In this section, we compare the proposed EA-CLIP with several state-of-the-art (SOTA) methods on three in-the-wild FER benchmarks, including AF-DB, AffectNet, and ExpW, as well as occlusion and variant-pose test sets to demonstrate the superiority of our approach.

4.3.1 Comparison on RAF-DB and AffectNet

Table 2 presents the results of various state-of-the-art (sota) methods evaluated on the RAF-DB and AffectNet datasets. Each method on the RAF-DB dataset is assessed based on its ability to recognize seven distinct expressions. From the table, we can find that our EA-CLIP achieves a significant improvement over existing approaches on the RAF-DB dataset, reaching a classification accuracy of 92.48%. Specifically, compared to CNN-based methods such as RAN [37], DMUE [40], Meta-Face2Exp [38], AMP-Net [16], MRAN [18], FG-AGR [17], and HAN-PFF [44], our method shows clear improvements ranging from 0.66% to 5.58%. While methods like MRAN focus on capturing multi-level relationships among local regions through a multi-relation aware

Table 2: Comparison results (%) with SOTA methods on RAF-DB and AffectNet (7/8 cls) datasets. The best results are in boldface, while the second-best results are underlined.

Method	Year	RAF-DB	AffectNet (7 cls)	AffectNet (8 cls)
SCN [12]	CVPR20	88.14	60.23	-
RAN [37]	TIP20	86.90	59.50	52.97
DMUE [40]	CVPR21	89.42	63.11	-
FDRL [45]	CVPR21	89.72	-	-
EfficientFace [71]	AAAI21	88.36	63.70	59.89
MA-Net [72]	TIP21	88.42	64.53	60.29
Meta-Face2Exp [38]	CVPR22	88.54	64.23	-
SOFT [73]	ECCV22	90.42	66.13	62.69
EAC [74]	ECCV22	89.99	65.32	-
PAT-Res101+attr [75]	TAC22	88.43	-	-
AMP-Net [16]	TCSVT22	89.25	64.54	61.74
MRAN [18]	TCSVT23	90.03	66.31	62.48
FG-AGR [17]	TCSVT23	90.81	64.91	60.69
HAN-PFF [44]	NN24	91.82	66.97	-
VTFF [19]	TAC21	88.14	64.80	61.85
TransFER [20]	ICCV21	90.91	66.23	-
APViT [15]	TAC22	91.98	66.91	-
MMATrans [49]	TII24	89.67	64.89	-
CLIPER [14]	ICME24	91.61	66.29	61.98
FER-former [64]	TMM24	91.30	-	-
CEPrompt [29]	TCSVT24	<u>92.43</u>	67.29	62.74
EA-CLIP (Ours)	2025	92.48	<u>67.07</u>	<u>62.68</u>

network, and FG-AGR utilizes adaptive graph association reasoning for fine-grained local feature extraction, they often neglect global information. Moreover, when compared with ViT-based methods such as VTFF [19], TransFER [20], APViT [15], and MMATrans [49], our approach improves by 0.5% to 4.34% over these approaches. This improvement is attributed to our integration of both visual and textual information, facilitating a more comprehensive understanding of facial expressions. Furthermore, compared to CLIP-based approaches, such as CLIPER [14], FER-former [64], and CEPrompt, our method achieves an improvement of 0.05% to 1.18%, demonstrating that our approach not only learns expression-specific features but also retains the generalized knowledge from pre-training.

Similar results are observed on the AffectNet dataset, which consists a large collection of unconstrained facial expression images, making it more complex than other datasets. To thoroughly evaluate the effectiveness of our method, we tested it on two configurations: AffectNet (7cls), which uses the same seven expression categories as RAF-DB, and AffectNet (8cls), which includes an additional “contempt” expression. Our method achieves the highest accuracy of 67.07% on AffectNet (7cls) and 62.68% on AffectNet (8cls), surpassing all current sota methods except CEPrompt by 0.16% to 7.57% on 7cls and 0.2% to 9.71% on 8cls. Particularly, compared to CLIPER, our

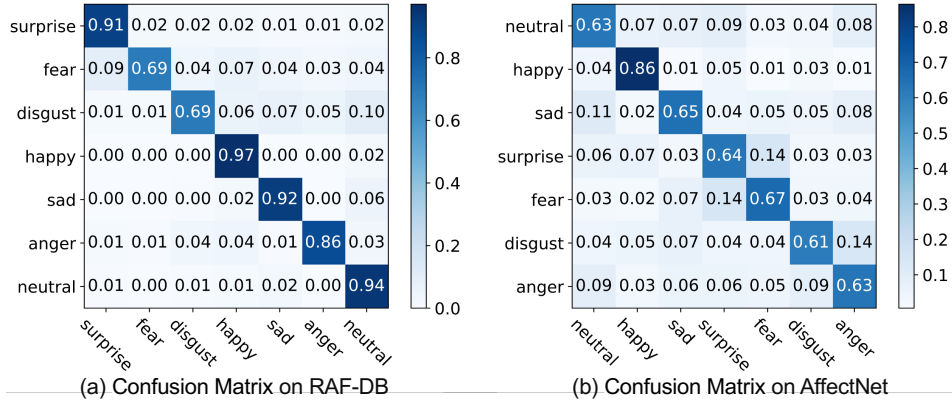


Fig. 2: Confusion matrices of EA-CLIP on (a) RAF-DB and (b) AffectNet.

method demonstrates significant improvements, highlighting its superior capability to handle complex expression recognition tasks. Although the recognition accuracy slightly trails CEPrompt on AffectNet, our EA-CLIP offers significant advantages in terms of efficiency. Specifically, CEPrompt’s two-stage fine-tuning process involves 36.11 GFLOPs and 32.29M trainable parameters, while our method implements an end-to-end single-stage fine-tuning manner, using only 42% of the computational resources and 46% of the trainable parameters. This efficiency makes our method more practical for real-world applications where computational resources may be limited.

Additionally, Fig. 2 illustrates the confusion matrix for EA-CLIP on the RAF-DB and AffectNet datasets. For RAF-DB (Fig. 2(a)), our approach achieves strong performance across most categories, with accuracy rates of 91%, 97%, 92%, 86%, and 94% for “surprise,” “happy,” “sad,” “anger,” and “neutral,” respectively. However, the uneven distribution of data in RAF-DB, where “fear” and “disgust” make up less than 10% of the dataset, results in increased confusion and reduced performance in these categories. Fig. 2(b) reveals that the AffectNet dataset poses greater recognition challenges than RAF-DB due to its complexity, stemming from a large number of unconstrained facial expression images. Our method achieves an optimal recognition rate of 86% for the “happy” expression in the AffectNet (7cls) dataset. However, the other six expressions have comparable performance levels, ranging from 61% to 67%. We attribute this to the difficulty in distinguishing “surprise,” “fear,” and “disgust,” which often results in mis-classification. Specifically, the “surprise” expression is frequently confused with “neutral” or “fear,” while “disgust” is often mistaken for “anger.”

4.3.2 Comparison on Occlusion and Variant-Pose Datasets

To further investigate the EA-CLIP method in more complex scenarios, we perform experiments on several face-occluded and pose-varied datasets, *i.e.*, Occlusion-RAF-DB, Occlusion-AffectNet, Pose-RAF-DB (Pose $\geq 30^\circ$ and Pose $\geq 45^\circ$), and Pose-AffectNet (Pose $\geq 30^\circ$ and Pose $\geq 45^\circ$) [37]. Tables 3 and 4 illustrate the experimental

Table 3: Comparison results (%) with SOTA methods on Occlusion-RAF-DB and Pose-RAF-DB datasets.

Method	Year	Occlusion	Pose $\geq 30^\circ$	Pose $\geq 45^\circ$
RAN [37]	TIP20	82.73	86.74	85.20
EfficientFace [71]	AAAI21	83.24	88.13	86.92
MA-Net [72]	TIP21	83.65	87.89	87.99
VTFF [19]	TAC21	83.95	87.97	88.35
AMP-Net [16]	TCSVT22	85.28	89.75	89.25
FG-AGR [17]	TCSVT23	88.15	91.02	90.50
MRAN [18]	TCSVT23	86.94	90.38	89.78
CEPrompt [29]	TCSVT24	<u>90.13</u>	<u>91.70</u>	91.35
EA-CLIP (Ours)	2025	90.15	91.78	<u>91.22</u>

Table 4: Comparison results (%) with SOTA methods on Occlusion-AffectNet and Pose-AffectNet datasets.

Method	Year	Occlusion	Pose $\geq 30^\circ$	Pose $\geq 45^\circ$
RAN [37]	TIP20	58.50	53.90	53.19
EfficientFace [71]	AAAI21	59.88	57.36	56.87
MA-Net [72]	TIP21	59.59	57.51	57.48
VTFF [19]	TAC21	62.98	60.61	61.00
AMP-Net [16]	TCSVT22	64.27	61.37	61.16
FG-AGR [17]	TCSVT23	64.24	61.26	61.15
MRAN [18]	TCSVT23	63.54	61.42	62.05
CEPrompt [29]	TCSVT24	<u>66.83</u>	64.28	63.44
EA-CLIP (Ours)	2025	66.89	<u>64.05</u>	<u>63.33</u>

results of EA-CLIP and several sota methods, with the experimental setup consistent with previous studies [18, 37]. From Table 3, it is evident that our EA-CLIP achieves superior recognition rates on Occlusion-RAF-DB, Pose-RAF-DB (Pose $\geq 30^\circ$) and Pose-RAF-DB (Pose $\geq 30^\circ$), which are 90.15%, 91.78% and 91.22%, respectively. Compared with FG-AGR [17] method, our method obtains the improvements of 2%, 0.76% and 0.72%, respectively. Similarly, as shown in Table 4, our EA-CLIP approach still obtains the highest accuracy on Occlusion-AffectNet (66.89%), while achieving comparable results to the current best method CEPrompt on Pose-AffectNet (Pose $\geq 30^\circ$, 64.05%), and Pose-AffectNet (Pose $\geq 45^\circ$, 63.33%). These experimental results verify the robustness of our proposed EA-CLIP method in handling occlusion and pose variations, benefiting from the integration of expression-specific vision-language knowledge and instance-specific visual information.

4.3.3 Comparison on ExpW

In Table 5, we compare the proposed method with several sora FER approaches on the ExpW dataset, renowned for its challenging features such as complex backgrounds

and multi-instance characteristics. Among them, SchiNet [69] achieves an accuracy of 73.10% by merging multiple datasets into a large-scale facial dataset for training, while APViT [15] slightly improves upon this by integrating a multi-branch local CNN with ViT to extract attention information from individual patches, reaching an accuracy of 73.48%. Compared to these methods, our EA-CLIP achieves the highest performance with an accuracy of 74.81%, surpassing APViT and CEPrompt by 1.33% and 0.44%, respectively. This demonstrates its superior ability to handle the complexities of the ExpW dataset.

Table 5: Comparison (%) with state-of-the-art methods on ExpW dataset.

Methods	Year	ExpW
DCN [11]	IJCV18	65.06
DCN+AP [11]	IJCV18	70.06
SchiNet [69]	TAC19	73.10
PAT-Res101 [75]	TAC22	69.29
PAT-Res101+attr [75]	TAC22	72.93
APViT [15]	TAC22	73.48
CEPrompt [29]	TCSVT24	74.37
EA-CLIP (Ours)	2025	74.81

4.4 Ablation Study

In this section, we perform ablation studies to assess the contributions of the key modules in EA-CLIP, specifically the Expression-aware Adapter (EAA) and the Instance-enhanced Expression Classifier (IEC), along with important hyper-parameters. All results are reported on the RAF-DB and AffectNet (7cls) datasets.

4.4.1 Effect of the Proposed Modules

To evaluate the effectiveness of our proposed key modules, we conduct ablation studies about the Expression-aware Adapter (EAA) and the Instance-enhanced Expression Classifier (IEC) on the RAF-DB and AffectNet datasets. From the results presented in Table 6, we can draw the following conclusions: (1) Utilizing the pre-trained CLIP model for zero-shot inference (Line 1, Eq. 1) results in poor performance, revealing the domain discrepancy between the pre-training task and the downstream FER task. (2) The baseline approach (Line 2) involves training the FER model from scratch on the respective datasets, initialized with parameters from the pre-trained CLIP model, and shows significant improvement across all datasets. Furthermore, we fine-tune the model with a specially designed text prompt (Line 3), *i.e.*, “a person with an expression of $[class_c]$ ”, surpassing the baseline by 0.95% on RAF-DB and 1.14% on AffectNet. However, the above full fine-tuning techniques require substantial computational resources and can lead to catastrophic forgetting of the general knowledge

acquired during pre-training. (3) Compared to the baseline, our EAA module (Line 3) enhances task-specific fine-tuning in a parameter-efficient manner, resulting in improvements of 2.08% on RAF-DB and 1.84% on AffectNet. This demonstrates the EAA module’s ability to capture expression-specific facial visual features while retaining the generalization knowledge of the pre-trained model. (4) Additionally, the IEC (Line 4) integrates instance-specific visual embeddings into the textual descriptors, enhancing the expressiveness of the textual embeddings and surpassing the baseline by 1.92% on RAF-DB and 1.58% on AffectNet. (5) Finally, the combination of EAA and IEC in our full model achieves superior performance over all variants, confirming the effectiveness of our proposed approach.

Table 6: Evaluation (%) of the key modules in EA-CLIP.

Modules	RAF-DB	AffectNet
Zero-shot	38.70	41.36
Fine-tune only	89.91	64.83
Fine-tune with prompt	90.86	65.97
EAA only	91.99	66.67
IEC only	91.83	66.41
EAA+IEC (Ours)	92.48	67.07

4.4.2 Impact of Text Prompts

As illustrated in CoOp [57], we acknowledge that prompts are crucial in downstream FER tasks as they can effectively bridge the gap between pre-training and downstream FER tasks, clearly showing that a well-designed prompt benefits FER.

To verify the influence of different prompt strategies, we conduct experiments on the RAF-DB and AffectNet datasets using various text prompts, as shown in Table 7. Initially, we set a baseline prompt that uses only category labels ($[class_c]$) without any specifically designed prompts, achieving an accuracy of 91.33% on RAF-DB and 66.09% on AffectNet. Moreover, we observe that different text prompts result in varying levels of performance on both datasets. For example, the prompt “a photo of $[class_c]$ ” surpasses the baseline prompt by 0.42% on RAF-DB and 0.37% on AffectNet. Additionally, the prompt “a person with an expression of $[class_c]$ ” showed the best performance, achieving 92.48% on RAF-DB and 67.07% on AffectNet. This empirical analysis substantiates that the selection of a prompt can significantly impact the performance of the FER model. The prompt “a person with an expression of $[class_c]$ ” offers the most effective guidance for the model to extract relevant information regarding facial expressions. As a result, we have designated “a person with an expression of $[class_c]$ ” as the default text prompt in all experiments.

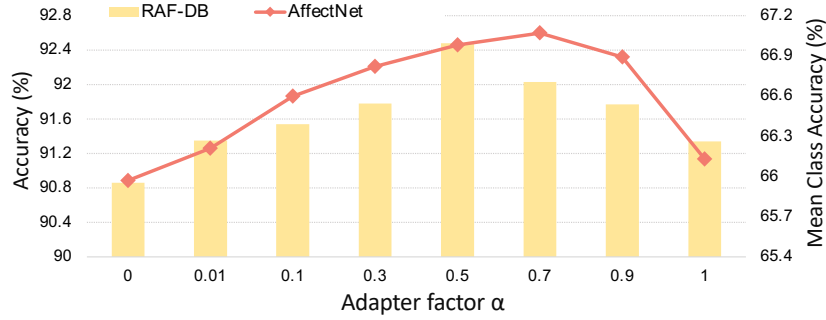
Table 7: Evaluation of (%) different text prompts.

Text Prompts	RAF-DB	AffectNet
$[class_c]$	91.33	66.09
a photo of $[class_c]$.	91.75	66.46
a photo of $[class_c]$ facial expression.	92.01	66.24
a person is $[class_c]$ in the photo.	91.66	66.28
a $[class_c]$ facial expression image.	92.05	66.83
a person with an expression of $[class_c]$.	92.48	67.07

4.4.3 Influence of hyper-parameters

We conduct ablation experiments to assess the impact of various hyper-parameters, including the adapter factor α in EAA, the angle factor γ in IEC, and the temperature factor τ . The results are illustrated in Fig. 3 and 4.

First, the **adapter factor α** serves as a trade-off to balance FER task-specific knowledge (*e.g.*, facial muscle movements) and general knowledge from the pre-trained CLIP model (*e.g.*, shapes, textures). Here, $\alpha=0$ represents the original transformer block without integrating any expression-specific knowledge. As shown in Fig. 3, a very small α leads to a significant performance drop due to the lack of task-specific knowledge, which is essential for understanding abstract and subtle facial expressions. Conversely, a very large α also disrupts the balance between general and specific features, resulting in suboptimal performance. Optimal performance is achieved with $\alpha=0.5$ on RAF-DB and $\alpha=0.7$ on AffectNet, with accuracies of 92.48% and 67.07%, respectively. This indicates that a suitable α enhances recognition performance while preserving the universal knowledge embedded in CLIP. Consequently, we set α to 0.7 for AffectNet and 0.5 for the other datasets.

**Fig. 3:** Influence of adapter factor α in Eq. 6 on RAF-DB and AffectNet datasets.

Second, the **angle factor γ** is a balancing scalar that determines the scale of the linear combination between instance visual embeddings and textual embeddings. As shown in Fig. 4, γ achieves optimal performance on the RAF-DB and AffectNet datasets at 0.9 and 0.7, respectively. A very small γ overly weakens the textual features,

reducing their impact on the final embeddings. Conversely, a very large γ neglects the instance-specific visual features, which are crucial for accurate facial expression recognition. Consequently, we set γ to 0.7 for AffectNet and 0.9 for the other datasets to maintain a balanced integration of textual and visual information.

Third, the **temperature parameter** τ regulates the smoothness of the probability distribution generated by the model. As observed in Fig. 4, $\tau=0.7$ yields the best results. This suggests that $\tau=0.7$ effectively balances the confidence of the model's predictions, enhancing its discriminative power without making the distribution too sharp or too flat. Hence, we set τ to 0.7 for all experiments.

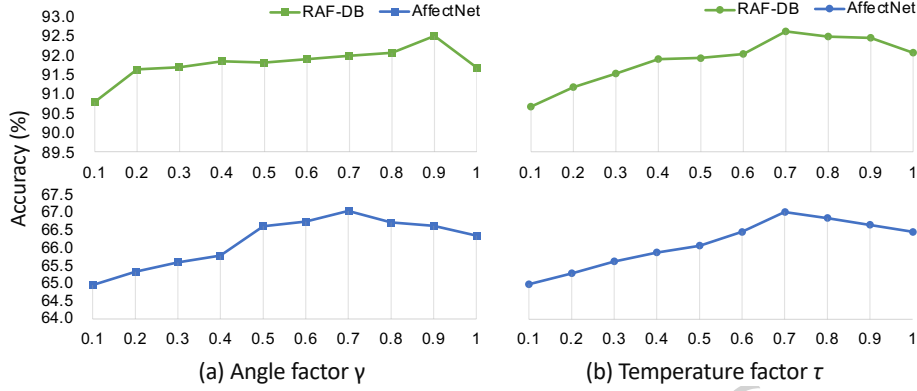


Fig. 4: Influence of angle factor γ in Eq. 9 and temperature factor τ in Eq. 7 on RAF-DB and AffectNet datasets.

4.4.4 Evaluation of Loss Functions

We further evaluate the performance of various loss functions, comparing the standard Cross-Entropy (CE) loss, Focal loss, Balance loss, Cosine loss, and Generalized Cross-Entropy loss. As illustrated in Table 8, the standard CE loss outperforms the others, achieving accuracies of 92.48% on the RAF-DB dataset and 67.07% on AffectNet. We hypothesize that the superior performance of the standard CE loss stems from its ability to effectively manage class imbalance in FER tasks. Unlike other loss functions that might overemphasize minority classes or require additional hyperparameter tuning, CE loss offers a straightforward optimization path that efficiently captures essential features for classification. In our experiments, we opt for the CE loss by default due to its consistent delivery of optimal performance across different datasets.

4.5 Robustness Analysis

To verify the robustness and reliability of our model, we conduct experiments across three trials, as presented in Table 9. From the table, it is evident that our method maintains consistent performance across various datasets. Besides, the performance metrics are averaged over three trials to ensure reliability and exhibit minimal variance,

Table 8: Evaluation of (%) different loss functions.

Loss	RAF-DB	AffectNet
Cross-Entropy	92.48	67.07
Focal Loss	91.86	66.84
Balance Loss	91.79	66.71
Cosine Loss	91.27	66.41
Generalized Cross-Entropy	91.11	66.29

indicating the stability of our approach. Specifically, the results show that the average accuracy on the RAF-DB dataset is 92.46%, with a very low standard deviation of ± 0.11 , suggesting strong consistency across trials. Similarly, for AffectNet (7cls and 8cls) and ExpW, the accuracy averages have standard deviations of ± 0.24 , ± 0.19 , and ± 0.09 , respectively. These results validate the robustness of our method under various experimental conditions.

Table 9: Evaluation of (%) model robustness under various seeds.

Seed	RAF-DB	AffectNet (7cls)	AffectNet (8cls)	ExpW
Seed1	92.48	67.07	62.68	74.81
Seed2	92.31	66.86	62.41	75.02
Seed3	92.59	67.33	62.79	74.73
Mean	92.46 ± 0.11	67.09 ± 0.24	62.63 ± 0.19	74.85 ± 0.09

4.6 Computational Complexity

To evaluate the computational complexity of EA-CLIP, we compared the inference time, floating point operations (FLOPs), and parameter count (Params) of our EA-CLIP on the AffectNet dataset. As shown in Table 10, our EAA module surpasses Fine-tune CLIP by 1.84% in performance while demonstrating significant computational efficiency. Specifically, Fine-tune CLIP requires substantial computational resources, with 86.19M trainable parameters, whereas EAA only tunes the last two transformer blocks along with expression-aware adapters, utilizing merely 17.2% of the parameters. Moreover, the IEC module exhibits less inference time and lower computational consumption than EAA, albeit with a slight decrease in recognition performance. Additionally, despite EA-CLIP (EAA+IEC) requiring more FLOPs than Fine-tune CLIP due to the additional computations in the instance-enhanced expression classifier, it achieves a performance gain of 2.24%. This analysis highlights EA-CLIP’s ability to maintain high recognition accuracy while effectively reducing computational usage.

Table 10: The inference time, number of flops (G) and parameters (M) obtained by different methods on the AffectNet dataset.

Method	Inference Time	FLOPs	Params.	AffectNet
Fine-tune CLIP	10.7s	11.62G	86.19M	64.83
EAA	10.7s	11.20G	14.82M	66.67
IEC	9.3s	4.11G	11.27M	66.41
EAA+IEC (EA-CLIP)	11.2s	15.31G	14.85M	67.07

4.7 Attention Visualization

To visually demonstrate the capacity of EA-CLIP comprehensively, we perform visualization of our proposed method through ScoreCAM [76] on several images, which are randomly selected from AffectNet dataset. As exhibited in Fig. 5, lighter colors indicate stronger ScoreCAM areas. From the figure, we observe that the vanilla CLIP model tends to focus on irrelevant regions that interfere with key facial muscles. Furthermore, the model frequently concentrates on non-essential areas such as facial edges, as evident in Columns 1, 3, 4, and 7 of Row 2. We hypothesize that this is due to the inherent object-centric bias in the pre-trained CLIP model, which leads it to prioritize global object-level features while neglecting the crucial fine-grained local features necessary for accurate expression recognition.

In contrast, our proposed expression-aware adapter module (EAA, Row 3) successfully directs attention to expression-related areas, such as the *Lips Part* in Column 4 and the *Lip Corner Depressor* in Column 7. Additionally, with the instance-enhanced expression classifier module (IEC, Row 4), our method effectively focuses on emotion-related representations, such as the *Lip Corner Puller* in Column 2 and the *Nose Wrinkler* in Column 7. By combining EAA and IEC (Row 5), our approach precisely targets critical expression areas while mitigating over-attention or under-attention to facial muscles, such as the *Lip Tightener* and *Lip Corner Depressor* in Column 3, as well as the *Inner Brow Raiser* and *Jaw Drop* in Column 5. These results demonstrate the superiority of task-specific fine-tuning in our EA-CLIP model, which effectively captures discriminative features associated with emotional expressions while preserving the general knowledge embedded in the pre-trained CLIP model. Additionally, our approach incorporates instance-specific visual features into the text, further enhancing recognition capabilities.

5 Conclusion

In this paper, we propose an emotion-aware adaptation framework, called EA-CLIP, for Facial Expression Recognition (FER), which integrates the pre-trained CLIP model to capture fine-grained emotional details and improving the accuracy and robustness of FER. We propose an Expression-aware Adapter to capture discriminative and emotion-specific visual representations through lightweight, task-specific fine-tuning, while maintaining the generalization capabilities of the CLIP model. Additionally, an Instance-enhanced Expression Classifier is designed to enhance the expressiveness of textual descriptors by incorporating instance-specific visual embeddings using

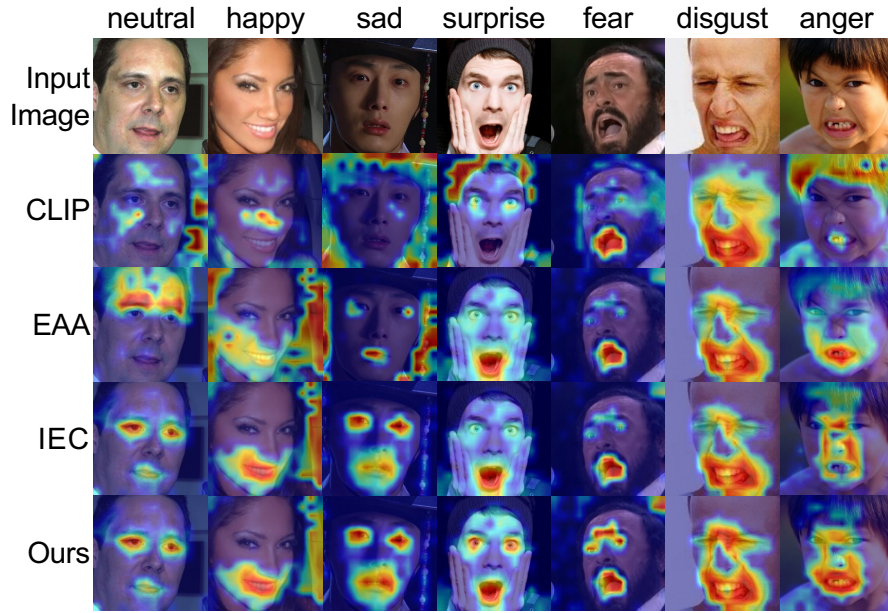


Fig. 5: Attention visualization [76] on AffectNet dataset.

spherical linear interpolation, resulting in a more precise and discriminative classifier. We conduct extensive experiments on three in-the-wild FER benchmarks, as well as occlusion and variant-pose test sets, demonstrating the superiority of our proposed approach over existing methods. Our results show that the integration of visual and textual information significantly enhances the model’s ability to capture emotion-related discriminative representations, leading to improved overall recognition performance. Future work will focus on further enhancing the adaptability of our framework to more complex and diverse FER scenarios, such as zero-shot and cross-domain settings. Additionally, we aim to explore the integration of additional modalities and advanced techniques to enhance FER.

Acknowledgments. The author(s) received no specific funding for this work.

Declarations

- **Competing interests:** The authors have no competing interests to declare that are relevant to the content of this article.
- **Author Contribution:** Conceptualization, methodology, J.H., M.L. and H.Z.; software, validation and data creation, J.H., M.L. and H.Z.; writing—original draft preparation, J.H., M.L. and H.Z.; writing—review and editing, J.H., M.L. and H.Z.; supervision, M.L. All authors have read and agreed to the published version of the manuscript.

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